Lecture III
Exploratory Spatial Data Analysis

STAT 679: Statistical Methods for Spatial Lattice Data-Spring 2026
University of Wisconsin-Madison

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Outline

1. Spatial Correlation in 1D
2. Simultaneous Autoregressive (SAR) Error Model
3. Conditional Autoregressive (CAR) Error Model
4. Moving Average (MA) Error Model
5. R-Demo
6. Assignment

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- We first consider modeling correlation in 1-dimensional (1D) space (i.e., along a transect).

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- A possible model for the spatially correlated errors is

$$\varepsilon_i = \rho\varepsilon_{i-1} + u_i, \quad i = 1, 2, \ldots, n$$

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$$\mathrm{Cor}(\varepsilon_i, \varepsilon_j) = \rho^{|i-j|},$$

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$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i,$$

$$\varepsilon_i = \rho \sum_{j=1}^{N} s_{ij} \varepsilon_j + u_i,$$

where $u_i \sim \text{ind } N(0, \sigma^2 v_i)$ and

$$s_{ij} = \begin{cases} 1, & i \neq j, \text{ and } i \text{ and } j \text{ are neighbors,} \\ 0, & i = j, \text{ or } i \text{ and } j \text{ are not neighbors.} \end{cases}$$

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$$N = \begin{pmatrix} s_{11} & s_{12} & \cdots & s_{1n} \\ s_{21} & s_{22} & \cdots & s_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ s_{n1} & s_{n2} & \cdots & s_{nn} \end{pmatrix}$$

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- The variance-covariance matrix of the error terms:

$$\Sigma = \sigma^2 \left[(I - \rho N)^T D^{-1} (I - \rho N) \right]^{-1}$$

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- The covariance of ε_i and ε_j is given by the (i, j)th element of Σ.
- In general, the correlations drop off as the ε_i's become more distant.
- Still, it is not immediately obvious what the individual elements of Σ are like, given a SAR model.

SAR Error Model in 2D (cont'd)

- R example: 3×4 lattice, first-order neighborhood, $\sigma^2 = 2.0$, and $\rho = 0.2$.
- The R code is reasonably accurate if the number of locations is small.
- Let $\sigma_{ij} = \mathrm{Cov}(\varepsilon_i, \varepsilon_j)$ and

$$\rho_{ij} = \mathrm{Cor}(\varepsilon_i, \varepsilon_j) = \frac{\sigma_{ij}}{\sqrt{\sigma_{ii}\sigma_{jj}}}.$$

- Note the relation between covariances and correlations in matrix form:

$$\begin{pmatrix} \frac{1}{\sqrt{\sigma_{11}}} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \frac{1}{\sqrt{\sigma_{nn}}} \end{pmatrix} \begin{pmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1n} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \cdots & \sigma_{nn} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{\sigma_{11}}} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \frac{1}{\sqrt{\sigma_{nn}}} \end{pmatrix}$$

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$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad \varepsilon_i \sim \text{normal distribution},$$

$$E(\varepsilon_i \mid \varepsilon_j,\ j \neq i) = \rho \sum_{j=1}^{n} c_{ij}\varepsilon_j,$$

$$\mathrm{Var}(\varepsilon_i \mid \varepsilon_j,\ j \neq i) = \sigma^2 v_i,$$

where

$$c_{ij} = \begin{cases} 1, & i \neq j, \text{ and } i \text{ and } j \text{ are neighbors,} \\ 0, & i = j, \text{ or } i \text{ and } j \text{ are not neighbors.} \end{cases}$$

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- The variance-covariance matrix of the error terms:

$$\Sigma = \sigma^2 (I - \rho N)^{-1} D$$

- The covariance of ε_i and ε_j is given by the (i, j)th element of Σ.

CAR Model in 2D (cont'd)

- In 1D space, CAR = SAR.
- Given an arbitrary $N = [c_{ij}]$ and ρ, the CAR model may not exist (e.g., Σ may not be symmetric or σ_{ii} may be negative).
- Any SAR model can be written in terms of a CAR model, but not vice versa. See for example an SAR(1) represented by a CAR(3) on p.409 Cressie.
- Not all CAR models can be written as SAR.
- Overall CAR models are more general and are sometimes preferred.

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MA Error Model in 2D (cont'd)

- One more model for 2D lattice data:

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$$\varepsilon_i = u_i + \rho \sum_{j=1}^{n} m_{ij} u_j,$$

where $u_i \sim$ ind $N(0, \sigma^2 v_i)$ and

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- Note: $m_{ii} = 0$.
- This model is known as the first-order moving average error model, or MA(1).
- The spatial weight m_{ij} represents the dependence of i on the neighbor j.

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MA Error Model in 2D (cont'd)

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- The covariance of ε_i and ε_j is given by the (i, j) element of Σ.
- MA models are not used as much as CAR or SAR models.

Outline

1. Spatial Correlation in 1D
2. Simultaneous Autoregressive (SAR) Error Model
3. Conditional Autoregressive (CAR) Error Model
4. Moving Average (MA) Error Model
5. R-Demo
6. Assignment

Ref: In-class demo and markdown files on canvas!

Outline

1. Spatial Correlation in 1D
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Homework Assignment #3

1. **(5 pts)** Refer to Figure: (*"Poverty by Socio-Economic Factors in Wisconsin/Illinois in 2000"*) from *Poverty in Wisconsin and Illinois Data Example Exploratory Data Analysis*. Interpret what you observe in the visualization for the other seven socio-economic factors.

2. **(10 pts)** Refer to *Poverty in Wisconsin and Illinois Data Example Simple Linear Regression*.
   (a) Complete the interpretation for all other regression coefficients for `m1`.
   (b) Complete the interpretation for all other regression coefficients for `m1.back`.
   (c) Perform the stepwise regression using `direction="both"`. How do the results change, if any?

Homework Assignment #3 (cont'd)

3. Consider the following grid of experimental plots:

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	25
26	27	28	29	30

The data in `corn.csv` consist of organic phosphorus in corn (y), organic phosphorus in soil (x_1), and inorganic phosphorus in soil (x_2) for each plot in the grid.
Consider the following models:

$$\text{Model I: } y = \beta_0 + \beta_1 x_1 + \varepsilon$$

$$\text{Model II: } y = \beta_0 + \beta_2 x_2 + \varepsilon$$

$$\text{Model III: } y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

(15 pts) For each model:

(a) Perform exploratory spatial data analysis on the data.
(b) Fit the models using the usual regression approach (ignoring spatial considerations). Construct and interpret a residual plot.
(c) Perform and interpret a test of significance for the slope(s) and intercept for the regression.

Homework Assignment #3 (cont'd)

4. **(5 pts)** For the three models in #3, test spatial correlation among the residuals, assuming a rook's case neighborhood and binary spatial weights, using `moran.test()`. Do this part for all 3 models.

   Hint:

   ```
   corn.lm <- lm(y ~ x1, data=corn)
   summary(corn.lm)
   plot(corn.lm$fitted, corn.lm$resid)
   moran.test(corn.lm$resid, mylw, alternative="two.sided")
   ```

5. **(5 pts)** A supposedly better test, although the math is complex, is implemented in `spdep` as follows:

   ```
   lm.morantest(corn.lm, mylw, alternative="two.sided")
   ```

   Perform this test for the three models in #3, and compare it with the results for #4. Do this part for all 3 models.

6. **(5 pts)** Interpret your findings for the problem as a whole using the likelihood ratio test. For the three models in #3, test either x_1 or x_2 by comparing a reduced model to the full model.

   Hint:

   ```
   anova(corn.lm_reduced, corn.lm_full, test="Chisq")
   ```